HK IMO Training 2018-2019
Problem Set 6
Date due: 3 November 2018

Problem 25. Point A lies on one ray and points B, C lie on the other ray of an angle with the vertex at O such that B lies between O and C. Let O_1 be the incentre of $\triangle OAB$ and O_2 be the center of the excircle of $\triangle OAC$ touching side AC. Prove that if $O_1A = O_2A$, then the triangle ABC is isosceles.

Problem 26. In $\triangle ABC$, the interior and exterior angle bisectors of $\angle BAC$ intersect the line BC at D and E, respectively. Let F be the second point of intersection of the line AD with the circumcircle of $\triangle ABC$. Let O be the circumcentre of $\triangle ABC$ and let D' be the reflection of D in O. Prove that $\angle D'FE = 90°$.

Problem 27. Let $a_0, a_1, \ldots, a_N$ be real numbers satisfying $a_0 = a_N = 0$ and

$$a_{i+1} - 2a_i + a_{i-1} = a_i^2$$

for $i = 1, 2, \ldots, N-1$. Prove that $a_i \leqslant 0$ for $i = 1, 2, \ldots, N-1$.

Problem 28. Determine the maximum size of a set of positive integers with the following properties:

(1) The integers consist of digits from the set $\{1, 2, 3, 4, 5, 6\}$.
(2) No digit occurs more than once in the same integer.
(3) The digits in each integer are in increasing order.
(4) Any two integers have at least one digit in common (possibly at different positions).
(5) There is no digit which appears in all the integers.